Algebra, Geometry, and Advanced Calculus II

Complete Handwritten Lecture Notes & Solved Problems

Part II: Algebra, Geometry, and Advanced Calculus II

Complex numbers, complex plane topology, limits, continuity, Cauchy-Riemann equations, differentiation, and analytic functions.

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Chapter 5. Complex Variable Functions and Calculus

1. Complex Plane

The set of complex numbers $\mathbb{C}$ forms a field under the operations of addition, subtraction, multiplication, and division. The zero element is $z = 0$, and the unit element is $z = 1$. The complex plane $\mathbb{C}$ can be considered as a two-dimensional vector space over the real numbers $\mathbb{R}$, with a basis of $\{1, i\}$. Alternatively, it can be viewed as a one-dimensional vector space over the complex numbers $\mathbb{C}$, with a basis of $\{1\}$.

If we let $(0, 1) = i$ and $(1, 0) = 1$, then any complex number $z = x + iy \in \mathbb{C}$ can be expressed as $z = x(1, 0) + y(0, 1) = x + iy$. The real part $x = \mathrm{Re}(z)$ and the imaginary part $y = \mathrm{Im}(z)$. Also, $i^2 = -1$.

2. Complex Functions

A complex function $f : \Omega \subset C \to C$ maps points in the complex plane to other points in the complex plane. If $z = x + iy$, then $w = u + iv$, where $u = u(x, y)$ and $v = v(x, y)$. The function can also be written as $w = f(z)$.

The function f can be decomposed into two component functions:

- $u = u(x, y)$: the real part function
- $v = v(x, y)$: the imaginary part function

A complex function can have multiple values: e.g.,
$w = \sqrt[n]{z} = (x^2 + y^2)^{\frac{1}{2n}} [\cos \frac{\theta + 2k\pi}{n} + i \sin \frac{\theta + 2k\pi}{n}]$, where $k = 0, 1, \ldots, n - 1$.

3. Limits and Continuity

For a function $w = f(z)$, if the real part and imaginary part of f have limits at a point, then the function is continuous at that point. The limit of $w = f(z)$ at z_0 is defined as:

$$\Delta w = f(z_0 + \Delta z) - f(z_0)$$

where $\Delta z = \Delta x + i\Delta y$, and $a = \alpha + i\beta$. The function f is said to be differentiable at z_0 if:

$$\Delta w = a\Delta z + o(\Delta z)$$

where $o(\Delta z)$ represents a term that is higher order than Δz.

4. Differentiation of Complex Functions

The derivative of a complex function f at z_0 is defined as:

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z}$$

where $\Delta w = a\Delta z + o(\Delta z)$. The function f is said to be differentiable at z_0 if this limit exists.

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The derivative of a complex function f(z) at z_0 is defined as the limit of the difference quotient as Δz approaches zero. The function can be expressed as a sum of its real and imaginary parts, u and v, respectively. The change in the function Δw is given by the sum of the real part of the change in u and the imaginary part of the change in v. The change in u and v can be expressed as linear combinations of Δx and Δy, with coefficients α and β. The derivative of the function is then given by the sum of the real and imaginary parts of the derivative of u and v, respectively. The function is differentiable at z_0 if and only if u and v are differentiable at (x_0, y_0) and satisfy the Cauchy-Riemann equations. The integral of a complex function f(z) along a path L is given by the sum of the real and imaginary parts of the integrals of u and v along the path. The integral can be expressed as the sum of the real and imaginary parts of the integrals of u and v along the path. 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- Analytic Functions - Extension of Elementary Real Functions

$\omega = f(z)$ is defined in $\Omega \subset C$. z_0 is an interior point of Ω. If there exists $r > 0$ such that $f(z)$ is differentiable in $B(z_0, r)$, then $f(z)$ is analytic at z_0.

Functions analytic in Ω:

- Analytic function ⇔ partial derivatives ($\partial u/\partial x$, $\partial u/\partial y$, $\partial v/\partial x$, $\partial v/\partial y$) are continuous in Ω.
- Points in Ω where $f(z)$ is not analytic: singular points.

$f(x)$ is a real-valued function defined on $I \subset R$. $F(z)$ is a complex-valued function defined in $\Omega \subset C$. If $I \subset \Omega$, $x \in I$, $f(x) = F(x)$, and $F(z)$ is analytic in Ω, then $F(z)$ is an analytic extension of $f(x)$.

- Polynomials: $P(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0$. In C, the analytic extension is:

$P(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0$

- Trigonometric and Hyperbolic Functions:

$P(x) = \cos(x) = (e^{(ix)} + e^{(-ix)})/2$, $Q(x) = \sin(x) = (e^{(ix)} - e^{(-ix)})/2i$

In C, the analytic extension is:

$P(z) = \cos(z)$, $Q(z) = \sin(z)$.

$\sin(z) = (e^{(iz)} - e^{(-iz)})/2i = (e^{(ix-y)} - e^{(-ix+y)})/2i = (\cos(x) + i \sin(x) - e^y) / 2i$

$= -i \cos(x) + i \sin(x) - e^y / 2i \cos(x) - 2i \sin(x)$

$= 1/2e^{(ix)} (\cos(x) - i \sin(x)) + e^y / 2 (\cos(x) + i \sin(x))$

$= (1/2e^{(ix)} + e^y / 2) \cos(x) - i (e^y / 2 - 1/2e^{(ix)}) \sin(x)$

= 1/2e^(ix) (cos(x) + i sin(x)) + e^y / 2 (cos(x) - i sin(x))

= 1/2 (e^(ix) - e^(-ix)) = 1/2 (e^(ix) - e^(-ix)) = cos(z).

∴ d/dz (cos(z)) = (1/2e^(ix) + e^y / 2) cos(x) - i (e^y / 2 - 1/2e^(ix)) sin(x)

= 1/2e^(ix) (cos(x) + i sin(x)) + e^y / 2 (cos(x) - i sin(x))

= 1/2 (e^(ix) - e^(-ix)) = 1/2 (e^(ix) - e^(-ix)) = sin(z).

Similarly, d/dz (cos(z)) = sin(z).

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The hyperbolic sine and cosine functions are also analytic on the complex plane:

$$\sinh z = \frac{e^z - e^{-z}}{2}, \quad \cosh z = \frac{e^z + e^{-z}}{2}$$

3. Root Functions:

$$\omega_k = n\sqrt{z} = n\sqrt{|z|} \cdot e^{i\frac{\arg z + 2k\pi}{n}}, \quad k = 0, 1, 2, \ldots, n-1$$

If the argument range is appropriately restricted, the multi-valued function can be "single-valued".

- Logarithmic Functions:

$$\omega = \ln z \quad \text{where} \quad z = re^{i\theta}, \quad \omega = u + iv$$

$$e^{u+iv} = re^{i\theta} = e^{u+iv}$$

$$\therefore u = \ln r, v = \theta + 2k\pi$$

$$\therefore \ln z = (\ln r + i(\theta + 2k\pi)) = \ln |z| + i \arg z$$

The principal branch of the logarithm is:

$$\ln z = \ln |z| + i \arg z + i2k\pi, \quad \text{k is a fixed positive integer}$$

The logarithmic function is multi-valued.

In the cut complex plane, the logarithmic function is locally analytic.

$$\frac{d(\ln z)}{dz} = \frac{d(\ln(x^2 + y^2) + i \arctan \frac{y}{x})}{dz} = \frac{x}{(x^2 + y^2)} - \frac{iy}{(x^2 + y^2)^2} = \frac{1}{z}$$

- Integration of Analytic Functions:

The complex function $f(z) = u(x, y) + iv(x, y)$ in the complex plane can be integrated along a simple closed path γ from A to B as the sum of two line integrals:

$$\int_{\gamma(A)}^{(B)} f(z)dz = \int_{\gamma(A)}^{(B)} udx - vdy + i\int_{\gamma(A)}^{(B)} vdx + udy$$

Cauchy's Integral Theorem:

If $f(z)$ is analytic in the interior of a simple closed path γ and continuous on γ, then:

$$\oint_{\gamma} f(z)dz = 0$$

This theorem is similar to Green's theorem, where the partial derivatives satisfy:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

This is the Cauchy-Riemann condition.

Cauchy Integral Formula:

Let the region D have boundary ∂D. If $f(\xi)$ is analytic in D and continuous in $D + \partial D$, then $f(\xi) = 1/(2\pi i) \int(D) f(\xi) / (\xi - z) d\xi$.

Proof: $\forall z \in D$, let $F(\xi) = f(\xi) / (\xi - z)$. Then $F(\xi)$ is analytic in D except at z.

Consider a circle Lp centered at z with radius $\rho > 0$. By Cauchy's integral theorem in the complex plane, $\int(\partial D) f(\xi) / (\xi - z) d\xi + \int(Lp) f(\xi) / (\xi - z) d\xi = 0$.

$\therefore \int(\partial D) f(\xi) / (\xi - z) d\xi = \int(Lp) f(\xi) / (\xi - z) d\xi$.

To prove the original formula, we need to show $\int(Lp) f(\xi) / (\xi - z) d\xi = 2\pi i f(z)$.

Let $\xi - z = \rho e^{(i\theta)}$, $\theta: 0 \sim 2\pi$.

Then $\int(Lp) d\xi / (\xi - z) = \int(0 \text{ to } 2\pi) i\rho e^{(i\theta)} / \rho e^{(i\theta)} d\theta = i \int(0 \text{ to } 2\pi) d\theta = 2\pi i$.

$\therefore | \int(Lp) f(\xi) / (\xi - z) d\xi - 2\pi i f(z) | = | \int(Lp) f(\xi) / (\xi - z) d\xi - \int(Lp) f(z) / (\xi - z) d\xi |$

$= | \int(Lp) (f(\xi) - f(z)) / (\xi - z) d\xi |$

$\because \forall \varepsilon > 0, \exists \delta > 0$, such that $0 < \rho < \delta$, $| f(\xi) - f(z) | < \varepsilon$.

$\therefore | \int(Lp) f(\xi) / (\xi - z) d\xi - 2\pi i f(z) | < \varepsilon$.

Thus, the function $f(\xi)$ in region D can be uniquely determined by its values on the boundary ∂D.

Furthermore, $f^{(n)}(z) = n! / (2\pi i) \int(\partial D) f(\xi) / (\xi - z)^{(n+1)} d\xi$, $z \in D$, $n = 1, 2, 3$.

By Cauchy's integral formula,

$f(z + \Delta z) - f(z) / \Delta z = 1/2\pi i \int(\partial D) f(\xi) / (\xi - z - \Delta z)(\xi - z) d\xi$

$= 1/2\pi i \int(\partial D) f(\xi) / (\xi - z - \Delta z)(\xi - z) d\xi$

$\therefore \left| f(z + \Delta z) - f(z) / \Delta z - 1/2\pi i \int(\partial D) f(\xi) / (\xi - z - \Delta z)(\xi - z)\, d\xi \right|$

$= 1/2\pi i \left| \int(\partial D) f(\xi) / (\xi - z - \Delta z)(\xi - z)\, d\xi - 1/2\pi i \int(\partial D) f(\xi) / (\xi - z - \Delta z)(\xi - z)\, d\xi \right|$

$= 1/2\pi i \left| \int(\partial D) f(\xi) / (\xi - z - \Delta z)(\xi - z)\, d\xi - 1/2\pi i \int(\partial D) f(\xi) / (\xi - z - \Delta z)(\xi - z)\, d\xi \right|$

$= 1/2\pi i \left| \int(\partial D) f(\xi) / (\xi - z - \Delta z)(\xi - z)\, d\xi - 1/2\pi i \int(\partial D) f(\xi) / (\xi - z - \Delta z)(\xi - z)\, d\xi \right|$

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$= 1/2\pi i \left| \int(\partial D) f(\xi) / (\xi - z - \Delta z)(\xi - z)\, d\xi - 1/2\pi i \int(\partial D) f(\xi) / (\xi - z - \Delta z)(\xi - z)\, d\xi \right|$

$= 1/2\pi i \left| \int(\partial D) f(\xi) / (\xi - z - \Delta z)(\xi - z)\, d\xi - 1/2\pi i \int(\partial D) f(\xi) / (\xi - z - \Delta z)(\xi - z)\, d\xi \right|$

$= 1/2\pi i \mid \int(\partial D) f(\xi) / (\xi - z - \Delta z)(\xi - z)\, d\xi - 1/2\pi i$

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The text and formulas in the image are as follows:

In ∂D, $|f(\xi)|$ is bounded. Let M be the upper bound, $\xi \in \partial D$. Let d be the shortest distance from z to ∂D. Then when $\xi \in \partial D$, $|\xi - z| \geq d > 0$. And when $|\Delta z| < d/2$, $|\xi - z - \Delta z| \geq |\xi - z| - |\Delta z| \geq d/2$. Thus, for $\varepsilon > 0$, take $\delta = \min\{d/2, \varepsilon\pi d^2/M\}$. Let L be the length of ∂D. When $|\Delta z| < \delta$, the original expression $\leq \delta/2\pi * M/L^2 * \int |\xi|\, d\xi \leq \varepsilon$. Therefore, $f'(z) = \lim(\Delta z \to 0)\ [f(z + \Delta z) - f(z)] / \Delta z = 1/2\pi i * \int f(\xi) / (\xi - z)^2\, d\xi$.

Assuming the result holds for n, then for n+1 (the result was proven for n=1 in the previous section).

$f^{(n+1)}(z) = [f^{(n)}(z)]'$

$= n! / 2\pi i * \int [f(\xi) / (\xi - z - \Delta z)^{(n+1)} - f(\xi) / (\xi - z)^{(n+1)}]\, d\xi$

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$= n! / 2\pi i * \int [f(\xi) / (\xi - z)^{(n+1)}] * [1 - (n+1)\Delta z / (\xi - z)]\, d\xi$

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$= n! /$

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- Series convergence necessary condition: $\lim_{n\to\infty} u_n = 0$

Series convergence sufficient condition: $S_n \leq M$ (bounded)

- Tests

- Cauchy test: $\forall \epsilon \in N \forall n > N, \sum_{k=n}^{n+p} u_k < \epsilon$, sufficient

- Comparison: $\exists N\ \forall n > N\ u_n \leq u_n$ series diverges, series converges

- D'Alembert's ratio test: $\exists N\ \forall n > N\ \frac{u_{n+1}}{u_n} < 1$ converges, > 1 diverges

Ratio test limit form: $\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = q\ q < 1$ converges, $q > 1$ diverges

- Cauchy root test: $\exists N\ \forall n > N\ \sqrt[n]{u_n} \leq 1$ converges, > 1 diverges

Root test limit form: $\lim_{n\to\infty} \sqrt[n]{u_n} = 1$ \$01\$ diverges

- Integral test: $f(x)[1, +\infty)$ non-decreasing, then $\sum f(n) = \sum u_n$ converges ($\sum u_n < \int_1^{\infty} f(x)dx)$

- Raabe test: $\lim_{n\to\infty} n\left(1 - \frac{u_{n+1}}{u_n}\right) = r\ r > 1$ converges, $r < 1$ diverges

Essentially compares with $\sum \frac{1}{n^r}$

- Kummer test: Suppose $\sum u_n$ is a positive series, $c_1, c_2, \ldots, c_n$ is a sequence of positive numbers

Let $K_n = c_n - c_{n+1}\frac{u_{n+1}}{u_n}$, if $\exists N, A > 0, \forall n > N, K_n \geq A$ then $\sum u_n$ converges, $K_n \leq 0$ then $\sum u_n$ diverges

Proof:

- $K_n = c_n - c_{n+1}\frac{u_{n+1}}{u_n} \geq A > 0$
- $\sum c_n u_n$ is decreasing and has a lower bound, so $c_n u_n$ has a limit

$\sum (c_n u_n - c_{n+1} u_{n+1}) = \sum (c_n u_n - c_{n+1} u_{n+1})$ converges

$\sum Au_n = A \sum u_n$ converges, so $\sum u_n$ converges

- $K_n = c_n - c_{n+1}\frac{u_{n+1}}{u_n} \leq 0$ then $u_n \geq \frac{1}{c_n}$ when $\sum \frac{1}{c_n}$ diverges, $\sum u_n$ diverges

Example: $c_n = n, \sum \frac{1}{n}$ diverges, use Raabe test

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- Alternating Series' Leibniz Test Method:
- If $(-1)^{(n-1)}U_n$ ($U_n > 0$) and $U_n \geq U_{n+1}$ and $U_n \to 0$, then $\sum(-1)^{(n-1)}U_n$ converges.
- (Can also prove convergence of $\sum U_n$ and $\sum(-1)^{(n-1)}U_n$ by Cauchy's convergence test).

Absolute Convergence: $\sum|U_n|$ converges implies $\sum U_n$ converges (U_n and $\sum U_n$ are alternating series).

Conditional Convergence: $\sum U_n$ converges, $\sum|U_n|$ diverges. ($\sum U_n$ and $\sum|U_n|$ both diverge)

- The rearrangement of absolutely convergent series is also absolutely convergent, with the same sum S. Using D'Alembert and Cauchy's direct method, the rearrangement of conditionally convergent series can be equal to any real number (Riemann).

- If $U_n = U$ and $V_n = V$, then $\sum U_n$ and $\sum V_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n =$ $\sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n =$ $\sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n =$ $\sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n =$ $\sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n =$ $\sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n =$ $\sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n =$ $\sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n =$ $\sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n =$ $\sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n =$ $\sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n =$ $\sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n = \sum U_nU_n =$

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∑U_n

The series

$$\sum_{n=0}^{\infty} \frac{x^{5n}}{(5n)!}$$

can be solved by recognizing that

$$I(x) = I(x)$$

, solving this differential equation, and substituting the boundary condition

$$I(0) = 1$$

,

$$I^{(n)}(0) = 0$$

for

$$n = 1, 2, \ldots, 5 - 1$$

. This yields the closed-form solution for the series.

The text discusses the properties of a function

$$f(x)$$

, where

$$f(x)$$

is a polynomial of degree

$$n$$

, and

$$f(x)$$

is continuous and differentiable. The function

$$f(x)$$

is defined as

$$f(x) = x^2 + 1$$

. The text also mentions that

$$f(x)$$

is a quadratic function, and it is continuous and differentiable everywhere. The text further discusses the properties of the function

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, such as its derivative and integral. The text also mentions that the function

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- Analytic Functions and Power Series Expansion

Analytic functions ↔ interior points are differentiable ↔ Cauchy-Riemann conditions / Cauchy integral theorem / power series (Laurent series & Taylor series).

Complex series: $\sum\infty k=1$ ak

A ≤ 0, ∃N, n>N when |Sn - S(= $\sum\infty k=1$ ak)| < ε

A ≤ 0, ∃N, A ≤ p, p>N when |ap - ap-1| < ε

Complex series: If D is a point in the complex plane, fk(z) defined in D, k=1, 2, ...

A ∊ D, series $\sum\infty k=1$ fk(z) = f1(z) + ... * fk(z) converges, then $\sum\infty k=1$ fk(z) in D converges, and the function f(z) = $\sum\infty k=1$ fk(z)

If the series converges in D for z, ∀ε > 0, ∃N(ε), it does not depend on z. When n > N(ε), ∀z ∊ D, then f(z) in D converges.

If each term fk(z) in D is continuous, then the function f(z) is also continuous in D;

If each term fk(z) in D is continuous, then ∫f(z)dz = $\sum\infty k=1$ ∫fk(z)dz;

If each term fk(z) in D is continuous, then f(z) is also continuous in D. The derivative f'(z) = $\sum\infty k=1$ f'(k)(z)

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Delta Laurent Series / Double Laurent Series

Each complex power series $\sum Cn(z-a)^n$ has a convergence circle $|z-a|=R$.

In the interior of the convergence circle, the series converges absolutely.

In the exterior, the series diverges.

On the boundary, the series may converge or diverge.

$\forall 0<\varepsilon$

If $f(z) = \sum Cn(z-a)^n$ in $|z-a|$

The coefficients satisfy $Cp = f^{(p)}(z) / p!$

For $\sum Cn(z-a)^n$, let $s = 1/(z-a)$, then $\sum Cn(z-a)^n = C1s + C2s^2 + \ldots + Cn\text{-}1s^{(n-1)} + \ldots$ $|s| = |1/(z-a)| < 1/(0 < 1/(z-a) < \infty)$. For $|z-a| > r$, $\sum Cn(z-a)^n$ converges. Therefore, $\sum Cn(z-a)^n = \sum Cn(z-a)^n + \sum Cn(z-a)^n$ converges, the convergence domain is $r < |z-a| < R$.

Laurent's Theorem: In the annulus H: $r < |z-a| < R$ $(0 \le r < R < \infty)$ of a function $f(z)$ that can be expanded into a double Laurent series $f(z) = \sum Cn(z-a)^n$, where $Cn = \int (f(z) / (z-a)^{(n+1)})\, dz$, $n=0, \pm 1, \ldots$

L+ is the circle $|z-a|=r$ (r

Proof: $\forall z \in H$: $r < |z-a| < R$, there exists p1, p2, $r < p1 < p2 < R$. Let L+ be the circle $|z-a|=r$ (r

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By Cauchy's integral formula,

$$f(z) = \frac{1}{2\pi i}\oint_{\Gamma_2}\frac{f(\xi)}{\xi - z}d\xi - \frac{1}{2\pi i}\oint_{\Gamma_1}\frac{f(\xi)}{\xi - z}d\xi$$

1) For the first integral:

$$\frac{f(\xi)}{\xi - z} - \frac{f(\xi)}{(\xi - a) - (z - a)} = \frac{f(\xi)}{\xi - a}\cdot\frac{1}{1 - \frac{z-a}{\xi-a}}$$

When $\xi \in \Gamma_2, |\xi - a| = r_2 > |z - a|$, so $|\frac{z-a}{\xi-a}| < 1$

$$\therefore \frac{f(\xi)}{\xi - z} = \frac{f(\xi)}{\xi - a}\left(1 - \frac{z - a}{\xi - a}\right)^{-1} = \frac{f(\xi)}{\xi - a}\sum_{n=0}^{\infty}\left(\frac{z - a}{\xi - a}\right)^{n}$$

The Laurent series in $|\xi - a| = r_2$ converges uniformly, so we can integrate term by term,

$$\therefore \frac{1}{2\pi i}\oint_{\Gamma_2}\frac{f(\xi)}{\xi - z}d\xi = \frac{1}{2\pi i}\oint_{\Gamma_2}\sum_{n=0}^{\infty}(\xi - a)^{n}\frac{f(\xi)}{(\xi - a)^{n+1}}d\xi$$

$$= \sum_{n=0}^{\infty}(\xi - a)^{n}\cdot\frac{1}{2\pi i}\oint_{\Gamma_2}\frac{f(\xi)}{(\xi - a)^{n+1}}d\xi \quad \left(C_n = \frac{1}{2\pi i}\oint_{\Gamma_2}\frac{f(\xi)}{(\xi - a)^{n+1}}d\xi\right)$$

$$= \sum_{n=0}^{\infty} C_n(z - a)^{n}$$

- For the second integral:

$$\frac{f(\xi)}{\xi - z} = -\frac{f(\xi)}{(\xi - a) - (z - a)} = -\frac{f(\xi)}{z - a}\cdot\frac{1}{1 - \frac{\xi-a}{z-a}}$$

When $\xi \in \Gamma_1, |\xi - a| = r_1 < |z - a|$, so $|\frac{\xi-a}{z-a}| < 1$

$$\therefore \frac{f(\xi)}{\xi - z} = -\frac{f(\xi)}{z - a}\left(1 - \frac{\xi - a}{z - a}\right)^{-1} = -\frac{f(\xi)}{z - a}\sum_{n=1}^{\infty}\left(\frac{\xi - a}{z - a}\right)^{n}$$

The Laurent series in $|\xi - a| = r_1$ converges uniformly, so we can integrate term by term,

$$\therefore \frac{1}{2\pi i}\oint_{\Gamma_1}\frac{f(\xi)}{\xi - z}d\xi = -\frac{1}{2\pi i}\oint_{\Gamma_1}\sum_{n=1}^{\infty}\frac{1}{(z-a)^n}f(\xi)(\xi - a)^n d\xi = -\sum_{n=1}^{\infty}\frac{C_n}{(z-a)^n}$$

$\left(C_n = \frac{1}{2\pi i}\oint_{\Gamma_1}\frac{f(\xi)}{(\xi - a)^{n+1}}d\xi\right)$

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The Laurent series of a function f(z) is given by:

$$f(z) = \sum_{n=0}^{\infty} C_n (z-a)^n + \sum_{n=1}^{\infty} \frac{C_n}{(z-a)^n} = \sum_{n=-\infty}^{\infty} C_n (z-a)^n$$

For a function f(z) to be analytic in the annulus |z-a|

$$f(z) = \sum_{n=0}^{\infty} C_n (z-a)^n$$

where

$$C_n = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(s)}{(s-a)^{n+1}} ds = \frac{f^{(n)}(a)}{n!}, n = 0, 1, 2, \ldots$$

This is the Taylor series of f(z) in the annulus 0<|z-a|

Examples:

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}, |z| < +\infty$$

$$\sin z = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)!}, |z| < +\infty$$

$$\cos z = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{(2n)!}, |z| < +\infty$$

$$(1+z)^{-1} = \sum_{n=0}^{\infty} (-1)^n z^n, |z| < 1$$

The Laurent series of a function f(z) has a convergence radius r < |z-a| < R. In the annulus |z-a|=r, |z-a|=R, there is at least one singularity of f(z). Specifically, if f(z) can be Taylor expanded, the convergence radius is |z-a|

Based on the above theorem, we can determine the Laurent (Taylor) series convergence radius by analyzing the singularities of f(z).

m-order zero point: If f(z) is analytic in the annulus |z-a|

$$f(z) = C_m(z-a)^m + C_{m+1}(z-a)^{m+1} + ... = (z-a)^m[C_m + C_{m+1}(z-a) + ...]$$

$$= (z-a)^m \varphi(z), \varphi(z) is analytic at z=a, C_m \neq 0, a is a zero point of f(z) of order m. Then a is a zero point of f(z) of order m.

Laurent Series:

- Removable singularity: In $0 < |z - a| < R$, $f(z) = \sum\infty$ n=1 $Cn(z - a)n$. When $n = 1, 2, 3, \ldots$, $Cn = 0$.
- Example: sin(z)/z
- Pole of order m: When $n = m+1, m+2, \ldots$, $Cn = 0$, $Cm \neq 0$.
- Example: cos(z) - pole of order 1
- Principal part: f(z) = Cm(z - a)^{-m} + ... + C0 + C1(z - a) + ... + Cn(z - a)^n + ..., a is the principal part of the pole.
- Essential singularity: There are infinitely many Cm not equal to zero.
- Example: e^(1/z)

Residue:

- Suppose a is f(z) removable singularity. Then there exists $R > 0$, in $0 < |z - a| < R$, f(z) is analytic.
- f(z) = ∑∞ n=0 a(z - a)^n
- Where C1 = ∫_L+ f(z) dz. L+: $|z - a| = \rho$, $(0 < \rho < R)$.
- C1 is the residue of f(z) at z = a. By Cauchy's integral theorem, Res(f(z), z = a) is related to the integral.

Residue Theorem:

- L+ is a closed path, f(z) is analytic in the region D except for finite singularities a1, ..., an.
- If f(z) is continuous in the region {a1, ..., an}, then ∫_L+ f(z) dz = ∑_k=1^n ∫_L+ f(z) dz = 2πi ∑_k=1^n ∫_L+ f(z) dz = 2πi ∑_k=1^n Res(f(z), z = ak).

If a is f(z) pole of order m, then $f(z) = \phi(z) / (z - a)^m$. Where $\phi(z)$ is analytic at $z = a$ and $\phi(a) \neq 0$. Then $\mathrm{Res}(f(z), z = a) = C1 = \int_{L+} f(z)\, dz = 1/(2\pi i) \int_{L+} \phi(z) / (z - a)^m\, dz = \phi^{(m-1)}(a) / (m-1)!$.

Logarithmic Residue:

- $\int_{L+} f(z)\, dz / f(z)\, dz$ is the logarithmic residue of f(z).

- If a is f(z) n-th zero point, then $\mathrm{Res}(f(z), z = a) = n$.
- If b is f(z) m-th pole point, then $\mathrm{Res}(f(z), z = b) = -m$.

Rouche's Theorem: Let L be a closed path, and f(z) and ϕ(z) be analytic functions inside and on L. If |f(z)| > |ϕ(z)| on L, then inside L, f(z) and f(z) + ϕ(z) have the same number of zeros.

Proof:

$$f(z) + \phi(z) = f(z)\left[1 + \frac{\phi(z)}{f(z)}\right]$$

$$\Delta_L \mathrm{Arg}[f(z) + \phi(z)] = \Delta_L \mathrm{Arg}[f(z)] + \Delta_L \mathrm{Arg}\left[1 + \frac{\phi(z)}{f(z)}\right]$$

$$\Delta_L \mathrm{Arg}\left[1 + \frac{\phi(z)}{f(z)}\right] = 0$$

Thus,

$$\Delta_L \mathrm{Arg}[f(z) + \phi(z)] = \Delta_L \mathrm{Arg}[f(z)] = N(f, L) = \frac{1}{2\pi i}\oint_L \frac{f'(z)}{f(z)}dz$$

If f(z) has no poles inside L, then

$$\frac{1}{2\pi i}\oint_L \frac{f'(z)}{f(z)}dz = N(f, L) = n$$

$$= \mathrm{Res}\left[\frac{f'(z)}{f(z)}\right] = \frac{1}{2\pi i}\oint_L \frac{d}{dz}[\ln f(z)]dz$$

The argument of the logarithm reflects the change in the argument of f(z) when z traverses L. At point z, the real and imaginary parts of ln f(z) are respectively (ln |f(z)|, ϕ). After one full traversal, the real part remains ln |f(z)| and the imaginary part is ϕ. Thus,

$$\frac{1}{2\pi i}\oint_L \frac{f'(z)}{f(z)}dz = \frac{1}{2\pi i}\{[\ln|f(z)| + i\phi] - [\ln|f(z_0)| + i\phi_0]\}$$

$$= \frac{\phi - \phi_0}{2\pi}$$

$$= \frac{\Delta_L \mathrm{Arg}[f(z)]}{2\pi}$$

$$= N(f, L) - P(f, L)$$

If f(z) has no poles inside L, then

$$\frac{\Delta_L \mathrm{Arg}[f(z)]}{2\pi} = N(f, L)$$

Rouche's Theorem: Let L be a closed path, and f(z) and ϕ(z) be analytic functions inside and on L. If:

1. They are both analytic inside and on L.

2. On L, |f(z)| > |ϕ(z)|.

Then inside L, f(z) and f(z) + ϕ(z) have the same number of zeros.

Chapter 6. Ordinary Differential Equations

1. Second-Order Linear Ordinary Differential Equations

1.1 Structure of Second-Order Linear Homogeneous Equations

Given the second-order linear homogeneous equation $y'' + p(x)y' + q(x)y = 0$, if y_1 and y_2 are two solutions, then $\{y_1, y_2\}$ forms a basis of the solution space of the linear space (rank = 2). Thus, $y(x) = c_1y_1(x) + c_2y_2(x)$, where $y_1(x)$ and $y_2(x)$ are linearly independent. Otherwise, $W(y_1, y_2) = W(x) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix} = 0$.

1.2 Structure of Second-Order Linear Non-Homogeneous Equations

Given the second-order linear non-homogeneous equation $y'' + p(x)y' + q(x)y = f(x)$, if $z_1(x)$ and $z_2(x)$ are two solutions, then $z_1(x) + p(x)z_1(x) + q(x)z_1(x) = f(x)$ and $z_2(x) + p(x)z_2(x) + q(x)z_2(x) = f(x)$. Thus, $[z_1(x) - z_2(x)]'' + p(x)[z_1(x) - z_2(x)]' + q(x)[z_1(x) - z_2(x)] = 0$. Hence, $z_1(x) - z_2(x)$ is a solution to the homogeneous equation $y'' + p(x)y' + q(x)y = 0$. Thus, $z(x) = c_1y_1(x) + c_2y_2(x) + z_0(x)$, where $z_0(x)$ is a particular solution of the non-homogeneous equation.

1.3 Constructing Linearly Independent Solutions

Given y_1 is a non-zero solution of the homogeneous equation $y'' + p(x)y' + q(x)y = 0$, and y_2 is a linearly independent solution. Since $y_2 \neq c_1y_1$, we have $y_2 = c_1y_1 + c_2y_1'$. Thus, $y_2 = c_1y_1 + c_2y_1'$. Therefore, $y_2 = c_1y_1 + c_2y_1'$. Thus, $y_2 = c_1y_1 + c_2y_1'$. Thus, $y_2 = c_1y_1 + c_2y_1'$. Thus, $y_2 = c_1y_1 + c_2y_1'$. Thus, $y_2 = c_1y_1 + c_2y_1'$. Thus, $y_2 = c_1y_1 + c_2y_1'$. Thus, $y_2 = c_1y_1 + c_2y_1'$. Thus, $y_2 = c_1y_1 + c_2y_1'$. Thus, $y_2 = c_1y_1 + c_2y_1'$. Thus, $y_2 = c_1y_1 + c_2y_1'$. Thus, $y_2 = c_1y_1 + c_2y_1'$. Thus, $y_2 = c_1y_1 + c_2y_1'$. Thus, $y_2 = c_1y_1 + c_2y_1'$. Thus, $y_2 = c_1y_1 + c_2y_1'$. Thus, $y_2 = c_1y_1 + c_2y_1'$. Thus, $y_2 = c_1y_1 + c_2y_1'$. Thus, $y_2 = c_1y_1 + c_2y_1'$. Thus, $y_2 = c_1y_1 + c_2y_1'$. Thus, $y_2 = c_1y_1 + c_2y_1'$.

Thus, $y_2 = c_1 y_1 + c_2 y_1'$. Thus, $y_2 = c_1 y_1 + c_2 y_1'$. Thus, $y_2 = c_1 y_1 + c_2 y_1'$. Thus, $y_2 = c_1 y_1 + c_2 y_1'$. Thus, $y_2 = c_1 y_1 + c_2 y_1'$. Thus, $y_2 = c_1 y_1 + c_2 y_1'$. Thus, $y_2 = c_1 y_1 + c_2 y_1'$. Thus, $y_2 = c_1 y_1 + c_2 y_1'$. Thus, $y_2 = c_1 y_1 + c_2 y_1'$. Thus, $y_2 = c_1 y_1 + c_2 y_1'$. Thus, $y_2 = c_1 y_1 + c_2 y_1'$. Thus, $y_2 = c_1 y_1 + c_2 y_1'$. Thus, $y_2 = c_1 y_1 + c_2 y_1'$. Thus, $y_2 = c_1 y_1 + c_2 y_1'$. Thus, $y_2 = c_1 y_1 + c_2 y_1'$. Thus, $y_2 = c_1 y_1 + c_2 y_1'$. Thus, $y_2 = c_1 y_1 + c_2 y_1'$. Thus, \(y_2 = c_1 y_1 + c_2 y_1' \

The original equation becomes:

$$(2y_1 + p(x)y_1)C'(x) + y_1C''(x) = 0$$

Solving for $C'(x)$ gives:

$$C'(x) = \frac{1}{y_1^2}e^{-\int p(x)dx}$$

Thus:

$$C(x) = \int \frac{e^{-\int p(x)dx}}{y_1^2}dx$$

Therefore:

$$y_2 = y_1\int \frac{e^{-\int p(x)dx}}{y_1^2}dx$$

- Using the two linearly independent solutions of the homogeneous equation, construct a particular solution of the non-homogeneous equation. Let the particular solution be $z_0 = C_1(x)y_1(x) + C_2(x)y_2(x)$, where $C_1(x), C_2(x)$ are linearly independent. Then:

$$z_0' = C_1(x)y_1(x) + C_2(x)y_2(x) + C_1'(x)y_1(x) + C_2'(x)y_2(x)$$

Given $z_0' = 0$, we have:

$$z_0'' = C_1(x)y_1''(x) + C_2(x)y_2''(x) + C_1'(x)y_1''(x) + C_2'(x)y_2''(x)$$

Substituting z_0, z_0', z_0'' into the non-homogeneous equation gives:

$$C_1y_1 + C_2y_2 = f(x)$$

Since:

$$C_1(x)y_1(x) + C_2(x)y_2(x) = 0$$

Thus:

$$C_1'(x) = -\frac{y_2(x)f(x)}{W(x)}$$

$$C_2'(x) = \frac{y_1(x)f(x)}{W(x)}$$

Therefore:

$$z_0 = C_1(x)y_1(x) + C_2(x)y_2(x) = \int_{x_0}^{x} \frac{y_2(t)f(t)}{W(t)}dt$$

3) Solving the homogeneous equation:

$$y'' + py' + qy = 0$$

Assume $y = e^{\lambda x}$, then:

$$(\lambda^2 + p\lambda + q)e^{\lambda x} = 0$$

1) If $\Delta > 0$, $y = C_1e^{\lambda_1 x} + C_2e^{\lambda_2 x}$

2) If $\Delta < 0$, $\lambda_{1,2} = \alpha \pm \beta i$. Thus:

$$y = e^{\alpha x}(C_1 \cos(\beta x) + C_2 \sin(\beta x))$$

3) If $\Delta = 0$, $y_1 = e^{-\frac{p}{2}x}$, using the method of undetermined coefficients, we get:

$$y_2 = xe^{-\frac{p}{2}x}$$

$$\therefore y = ae^{-\frac{b}{2}x} + c_2xe^{-\frac{b}{2}x}$$

Solve higher-order equations in the same way.

(4) Solve non-homogeneous equations.

- Generally, $z_0 = \int_{x_0}^{x} \frac{y_1(t)y_1(x)+y_1(t)y_2(x)}{W(t)} \cdot f(t) \cdot dt.$

- $z = ay_1 + c_2y_2 + z_0.$

Specifically, if the non-homogeneous term $f(x) = P_n(x)e^{ax}$ (a is a complex number, P_n is a polynomial).

1) If μ is not a characteristic root of the homogeneous equation, then the non-homogeneous differential equation has the form:

$$z_0 = Q_n(x)e^{ax}$$

- If μ is a single characteristic root of the homogeneous equation, then:

$$z_0 = xQ_n(x)e^{ax}$$

3) If μ is a double characteristic root of the homogeneous equation, then:

$$z_0 = x^2Q_n(x)e^{ax}$$

Substitute the special solution $z_0(x)$ into the non-homogeneous equation. Through comparison of coefficients, determine $Q_n(x)$.

For $y'' + py' + qy = f(x), f(x) = Re[f_1(x) + f_2(x)]$ (where $f_1(x)e^{ax}$ also satisfies $P_n(x)e^{ax}$).

We need to solve the two equations:

$$f_1(x) = y'' + py' + qy$$

$$f_2(x) = y'' + py' + qy$$

The solution $z_1(x) + z_2(x)$ is the special solution of the non-homogeneous equation.

For the Euler equation $\sum_{i=0}^{n} a_i x^i y^{(i)} = 0$, by substituting $t = \ln x$, we can obtain a homogeneous equation with constant coefficients.

- First-order linear constant coefficient differential equations.

$$\dot{y} = Ay + f(x)$$

$$\dot{y} = (y_1, y_2, \ldots, y_n)^T, A(x) = (a_{ij}(x))_{n\times n}, f(x) = (f_1(x), f_2(x), \ldots, f_n(x))^T$$

Second-order linear constant coefficient differential equations are a special case of first-order linear differential equations.

- Structure of the solution of first-order linear differential equations.

The solution set of a first-order linear constant coefficient differential equation is a linear space of dimension n. The general solution of a non-homogeneous equation can be expressed as the general solution of the corresponding homogeneous equation plus a particular solution of the non-homogeneous equation.

- Solution of constant coefficient linear systems.

Given

$$\dot{y} = Ay + f(x)$$

, the corresponding homogeneous equation

$$\dot{y} = Ay$$

has the solution

$$y = Pe^{Ax}$$

. Substituting into

$$(A - \lambda E)P = 0$$

, we get

$$\lambda$$

is an eigenvalue of A, and P is an eigenvector of A.

- If the eigenvalues

  $$\lambda_1$$

  and

  $$\lambda_2$$

  are not equal, the corresponding eigenvectors

  $$\vec{p}_1$$

  and

  $$\vec{p}_2$$

  are linearly independent. The general solution of

  $$\dot{y} = Ay$$

  is

  $$\vec{p}_1 e^{\lambda_1 x}$$

  and

  $$\vec{p}_2 e^{\lambda_2 x}$$

  .

The general solution of

$$\dot{y} = Ay + f(x)$$

is

$$y = C_1 \vec{p}_1 e^{\lambda_1 x} + C_2 \vec{p}_2 e^{\lambda_2 x}$$

.

If the eigenvalues are complex conjugates, the corresponding eigenvectors are linearly dependent. The general solution of

$$\dot{y} = Ay$$

is

$$\vec{p}_1 e^{\lambda_1 x}$$

and

$$\vec{p}_2 e^{\lambda_2 x}$$

.

- If the eigenvalues

  $$\lambda_1 = \lambda_2$$

  , the general solution of

  $$\dot{y} = Ay$$

  is

  $$\vec{p}_1 e^{\lambda_1 x} + \vec{p}_2 x e^{\lambda_1 x}$$

  .

The general solution of

$$\dot{y} = Ay + f(x)$$

is

$$y = C_1 \vec{p}_1 e^{\lambda_1 x} + C_2 \vec{p}_2 x e^{\lambda_1 x}$$

.

For systems of more than two equations, the rules are similar.

Delta Constant Substitution Method:

For

$$\vec{y}' = A\vec{y} + f(x)$$

, let

$$\vec{y} = C(x)\vec{\phi}(x)$$

, substitute into the original equation to get:

$$\vec{\phi}'(x)C(x) + \vec{\phi}(x)C'(x) = A\vec{\phi}(x)C(x) + f(x)$$

.

$$\therefore \vec{\phi}'(x) = A\vec{\phi}(x)$$

$$\therefore \vec{\phi}'(x)C(x) = A\vec{\phi}(x)C(x)$$

$$\therefore \vec{\phi}(x)C'(x) = f(x)$$

.

$$\therefore C'(x) = \vec{\phi}^{-1}(x) \cdot f(x)$$

.

$$\therefore C(x) = \int_{x_0}^{x} \vec{\phi}^{-1}(x) \cdot f(x)\,dx + C$$

.

(Translation of the rest of the text is not provided as it is not in English and appears to be a continuation of the mathematical derivation.)

Fourier Analysis and Laplace Transform.

- Orthogonal Basis of Functions $\{1, \sin(n\pi x/L), \cos(n\pi x/L), \sin(2n\pi x/L), \cos(2n\pi x/L), \ldots\}$

Basis Norm $|e_n| = \sqrt{(1/L)} \int[-L, L] \cos^2(n\pi x/L)\, dx = \sqrt{(1/2)} \int[-L, L] (1 - \cos(2n\pi x/L))\, dx = \sqrt{(1/2)} \int[-L, L] 1\, dx = \sqrt{(1/2)}\, L$

$\therefore f(x) = (1/2) \int[-L, L] f(x)\, dx + (1/2) \int[-L, L] f(x) \cos(n\pi x/L)\, dx + \ldots + (1/2) \int[-L, L] f(x) \cos(n\pi x/L)\, dx + \ldots$

$= (a_0/2) + \sum[n=1,\infty] (a_n \cos(n\pi x/L) + b_n \sin(n\pi x/L))$ (at the continuity points).

Where $a_n = (1/L) \int[-L, L] f(x) \cos(n\pi x/L)\, dx$, $n=0,1,2,\ldots$

$b_n = (1/L) \int[-L, L] f(x) \sin(n\pi x/L)\, dx$, $n=1,2,\ldots$

- Fourier Integral Formula:

$f(x) = (a_0/2) + \sum[n=1,\infty] (a_n \cos(n\pi x/L) + b_n \sin(n\pi x/L))$

$= (1/2L) \int[-L, L] f(t)\, dt + \sum[n=1,\infty] (1/L) \int[-L, L] f(t) (\cos(n\pi x/L) \cos(n\pi x/L) + \sum[m=1,\infty] (\cos(n\pi x/L) \cos(m\pi x/L))\, dt$

$= (1/2L) \int[-L, L] f(t)\, dt + \sum[n=1,\infty] (1/L) \int[-L, L] f(t) \cos(n\pi x/L) (\cos(t) - t)\, dt$

Let $u_n = n\pi/L$, $\therefore u_n = u_n - u_{n-1} = \pi/L$.

$\therefore \sum[n=1,\infty] (1/L) \int[-L, L] f(t) \cos(n\pi x/L) (\cos(t) - t)\, dt = \sum[n=1,\infty] (1/L) \int[-L, L] f(t) \cos(u_n (x-t))\, dt$

$= (1/\pi) \sum[n=1,\infty] (1/L) \int[-L, L] f(t) \cos(u_n (x-t))\, dt\, \Delta u_n$

When $L \to +\infty$, $\Delta u_n \to 0$, so the above formula becomes $\sum[n=1,\infty] (1/\pi) \int[-L, L] f(t) \cos(u_n (x-t))\, dt\, \Delta u_n = \int[0, +\infty] \phi(u)\, du$

$\therefore f(x) = (1/\pi) \int[0, +\infty]\, du \int[-\infty, +\infty]\, f(t) \cos(u\,(x-t))\, dt \quad -\infty < x < +\infty.$

$= \int[0, +\infty]\, [a(u) \cos(ux) + b(u) \sin(ux)]\, du.$

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- Fourier Transform

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} du \int_{-\infty}^{+\infty} f(t)\cos u(x-t)dt$$

$$= \frac{1}{2\pi}\int_{-\infty}^{+\infty} du \int_{-\infty}^{+\infty} f(t)\cos u(x-t)dt \quad (\cos u(x-t) is u' s even function)$$

$$= \frac{1}{2\pi}\int_{-\infty}^{+\infty} du \int_{-\infty}^{+\infty} f(t)[\cos u(x-t) + i\sin u(x-t)]dt \quad (\sin u(x-t) is u' s odd function)$$

$$= \frac{1}{2\pi}\int_{-\infty}^{+\infty} du \int_{-\infty}^{+\infty} f(t)e^{iux}dt$$

$$= \frac{1}{2\pi}\int_{-\infty}^{+\infty} e^{iux}du \int_{-\infty}^{+\infty} f(t)e^{-itu}dt$$

$$= \frac{1}{2\pi}\int_{-\infty}^{+\infty} F(u)e^{iux}du$$

$$F(u) is the Fourier transform of f(x), F(u) = \int_{-\infty}^{+\infty} f(t)e^{-itu}dt, written as F(u) = F[f(x)]$$

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} F(u)e^{iux}du is the Fourier inverse transform, written as f(x) = F^{-1}[F(u)]$$

Differentiation property: If f(x) is absolutely integrable and continuous on (-∞, +∞), and its first derivative also satisfies the Dirichlet condition, then

$$F[f'(x)] = iuF[f(x)]$$

$$F[f'(x)] = \int_{-\infty}^{+\infty} f'(x)e^{-itu}dx = f(x)e^{-itu}\Big|_{-\infty}^{+\infty} + iu\int_{-\infty}^{+\infty} f(x)e^{-itu}dx$$

$$= iuF[f(x)]$$

$$\therefore F[f^{(n)}(x)] = (iu)^n F[f(x)]$$

$$F[\int_{-\infty}^{+\infty} f(x)dx] = \frac{1}{iu}F[f(x)]$$

$$\because \frac{d^n F(u)}{du^n} = F[-ix^n f(x)]$$

Convolution property: If f1(x) and f2(x) are absolutely integrable and satisfy the Dirichlet condition on (-∞, +∞), then

$$F[f_1(x) * f_2(x)] = F_1(u)F_2(u)$$

f1 and f2's convolution: f1(x) * f2(x) = ∫_{-∞}^{+∞} f1(y) f2(x-y) dy.

No.

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$$F[f_1(x)*f_2(x)] = \int_{-\infty}^{+\infty} [f_1(x)*f_2(x)]e^{-iux}dx$$

$$= \int_{-\infty}^{+\infty} [\int_{-\infty}^{+\infty} f_1(y)f_2(x-y)dy]\, e^{-iux}dx$$

$$= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f_1(y)e^{-iuy}f_2(x-y)e^{-iux}dydx$$

$$= \int_{-\infty}^{+\infty} f_1(y)e^{-iuy}dy \int_{-\infty}^{+\infty} f_2(x-y)e^{-iux}dx$$

$$= F_1(u)F_2(u)$$

$$\therefore f_1(x)*f_2(x) = F^{-1}[F_1(u)F_2(u)]$$

- Laplace Transform

Let f(t) be defined in [0, +∞) and have a Laplace transform in the region of convergence. Then the integral of f(t) e^(-pt) dt in the region of convergence determines a function F(p) of p.

$$F(p) = L[f(t)]$$

$$f(t) = L^{-1}[F(p)]$$

Differentiation Property: If f(t) and f'(t) both have Laplace transforms, and L[f(t)] = F(p), then

$$L[f'(t)] = pF(p) - f(0)$$

$$L[f'(t)] = \int_0^{+\infty} f'(t)e^{-pt}dt = f(t)e^{-pt}\Big|_0^{+\infty} + p\int_0^{+\infty} f(t)e^{-pt}dt$$

$$= -f(0) + pF(p), \quad \mathrm{Re}(p) > c$$

Generally, L[f^(n)(t)] = p^n F(p) - p^(n-1) f(0) - p^(n-2) f'(0) - ... - f^(n-1)(0)

If f(0) = f'(0) = ... = f^(n-1)(0) = 0, then

$$L[\int_0^t f(x)dx] = \frac{1}{p}F(p)$$

Let $\lambda(t) = \int_0^t f(\tau)\, d\tau$, then $\lambda(0) = 0$, $\lambda'(t) = f(t)$

By the properties of integration, $\int[f(t)] = \int[\lambda'(t)] = p \int[\int_0^t f(\tau)\, d\tau]$

$\therefore \int[\int_0^t f(\tau)\, d\tau] = 1/p \int[f(t)] = 1/p\, F(p)$.

- Inverse Laplace Transform

If the Laplace transform of $f(t)$ is $F(p)$, i.e., $\int[f(t)] = F(p)$, then

$f(t) = 1/2\pi i \int_{[\beta+i\infty]}^{(\beta-i\infty)} F(p)\, e^{(pt)}\, dp$, $t > 0$.

If $p_1, \ldots, p_n$ are all the poles of $F(p)$, and when $p \to \infty$, $F(p) \to 0$, then

$f(t) = 1/2\pi i \int_{[\beta+i\infty]}^{(\beta-i\infty)} F(p)\, e^{(pt)}\, dp = \sum_{(n=1 \text{ to } n)} \text{Res}[F(p)\, e^{(pt)}]$ at $p = p_n$

$\int[f_1(t) * f_2(t)] = \int_0^\infty [f_1(t) * f_2(t)]\, e^{(-pt)}\, dt$

$= \int_0^\infty [\int_0^t f_1(\tau)\, f_2(t-\tau)\, d\tau]\, e^{(-pt)}\, dt$

$= \int_0^\infty f_1(\tau)\, [\int_0^\infty f_2(t-\tau)\, e^{(-pt)}\, dt]\, d\tau$

$= \int_0^\infty f_1(\tau) \int_0^\infty f_2(s)\, e^{(-p(\tau+s))}\, ds\, d\tau$

$= \int_0^\infty f_1(\tau)\, e^{(-pt)}\, d\tau - \int_0^\infty f_2(s)\, e^{(-ps)}\, ds$

$= F_1(p)\, F_2(p)$.

Chapter 8. About the Completeness of the Real Number System

- **Bolzano's Theorem (Bolzano)**

A bounded sequence has a convergent subsequence.

- **Monotone Bounded Theorem**

A bounded monotone sequence converges.

- **Interval Nesting Theorem**

If the closed intervals $\{[a_n, b_n]\}$ satisfy the conditions:

(i) $[a_{n+1}, b_{n+1}] \subset [a_n, b_n]$, $n = 1, 2, \dots$

(ii) $\lim_{n \to \infty}(b_n - a_n) = 0$, then there is a unique point x that belongs to all the closed intervals $[a_n, b_n]\,(n = 1, 2, \dots)$.

- **Finite Covering Theorem**

From any open covering of $[a, b]$, we can extract a finite open covering of $[a, b]$.

- **Accumulation Point Theorem**

A bounded infinite set has at least one accumulation point.

- **Least Upper Bound Principle (Least Lower Bound Principle)**

Any bounded (non-bounded) set has an upper bound (lower bound).

- **Completeness Theorem (Cauchy)**

A sequence $\{a_n\}$ converges if and only if: $\forall \epsilon > 0, \exists N \in N^+, \forall n, m > N, |a_n - a_m| < \epsilon$

$6 \Rightarrow 2$: **Least Upper Bound** $\Rightarrow$ **Monotone Bounded**

Proof: Consider the decreasing case.

Given the sequence $\{a_n\}$ is decreasing and bounded, so $\{a_n\}$ has a lower bound, denoted $A = \inf\{a_n\}$.

$\therefore \forall n, a_n \geq A; \forall \epsilon > 0, \exists N \in N^+, a_N < A + \epsilon$

Since $\{a_n\}$ is decreasing, $\therefore \forall n > N, a_n \leq a_N < A + \epsilon$

$\therefore |a_n - A| < \epsilon$

$\therefore \lim_{n \to \infty} a_n = A.$

2→3: Monotone bounded → Interval套

Proof: Given {an} monotone increasing and {bn} monotone decreasing, and b1 is an upper bound of {an}, a1 is a lower bound of {bn}. By the monotone bounded theorem, {an} and {bn} both converge. Let lim an = A, lim bn = B. Therefore, lim (bn - an) = 0, so A = B. Thus, ∀n, an ≤ A; ∀ε > 0, ∃N, an > A - ε. Therefore, A = sup an, similarly B = inf bn. And ∀m, n, am ≤ bn. Thus, ∀n, bn is an upper bound of {an}, and an is a lower bound of {bn}. Therefore, ∀n, A ≤ bn, an ≤ B. Thus, ∀n, an ∈ [A, B]. Therefore, A ∈ [a1, b1].

Assume μ ∈ [a1, b1], then ∀n, an ≤ μ ≤ bn. By the property of limits, A ≤ μ ≤ A, so μ = A.

3→1: Interval套 → Bolzano (Bolzano)

Proof: Given {xn} bounded. ∴ ∃a1, b1, such that ∀n, xn ∈ [a1, b1]. Take any x1 ∈ [a1, b1], divide the interval [a1, b1] into [a1, (a1 + b1)/2] and [(a1 + b1)/2, b1]. Then among these, there must be a subsequence of {xn} that is unbounded. Take one of these intervals, [a2, b2], b2 - a2 = (b1 - a1)/2. Take any x2 ∈ [a2, b2], n2 > n1, ...

Thus, we generate the interval套 [a1, b1] (where [a1, b1] is half of [a1, b1]) and a subsequence {xn} that satisfies xn ∈ [a2, b2]. By the Bolzano定理, ∃! c ∈ [a1, b1], such that {xn} converges to c.

1→7: Compactness (Bolzano) → Completeness (Cauchy)

Proof: Necessity.

Assume $\{x_n\}$ converges. Then $\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n > N$ such that $|x_n - A| < \varepsilon/2$.

When $m, n > N$, then $|x_m - A| < \varepsilon/2$, $|x_n - A| < \varepsilon/2$, thus $|x_m - x_n| \le |x_m - A| + |x_n - A| < \varepsilon/2 + \varepsilon/2 = \varepsilon$.

Sufficiency.

Assume $\{x_n\}$ is a Cauchy sequence. Then for any $\varepsilon > 0$, $\exists N_1 > 0$, $\forall m, n > N_1$, such that $|x_m - x_n| < 1$.

$\therefore \forall n > N_1$, there is $|x_n - x_{n+1}| < 1$, $|x_n| < |x_{n+1}| + 1$.

So the sequence $\{x_n\}$ is bounded, and there exists a subsequence $\{x_{n_k}\}$, such that $\lim x_{n_k} = A$.

$\forall \varepsilon > 0$, $\exists N_2 > 0\ \forall k > N_2$, such that $|x_{n_k} - A| < \varepsilon/2$. $\exists N_3 > 0$, $\forall m, n > N_3$, such that $|x_m - x_n| < \varepsilon/2$.

Take $N = \max(N_2 + 1, N_3 + 1)$.

When $n > N$, there is $n > N > N_3$.

$\therefore |x_n - A| \le |x_n - x_{n_N}| + |x_{n_N} - A| < \varepsilon/2 + \varepsilon/2 = \varepsilon$.

$\therefore \lim x_n = A$.

3→4: Interval Covering → Finite Covering.

Proof: If not, then $[a, b]$ cannot be covered by a finite number of open intervals.

Divide the interval $[a, b]$ into two halves, at least one half cannot be covered by a finite number of open intervals.

Repeat this process on $[a_1, b_1]$ (if both halves are covered, take either one).

Repeat this process on $[a_1, b_1]$ into two halves, at least one half cannot be covered by a finite number of open intervals.

This process will yield a sequence of intervals $[a_k, b_k]$, such that $\exists c \in [a_k, b_k]$ for any $c \in [a_k, b_k]$.

$\therefore \lim (b_k - a_k) = 0$, $c \in [a_k, b_k]$ for any $c \in [a_k, b_k]$. $\therefore \exists k$ such that $[a_k, b_k] \subset c$, which contradicts the assumption of finite covering.

$\therefore \lim (b_k - a_k) = 0$, $c \in [a_k, b_k]$ for any $c \in [a_k, b_k]$. $\therefore \exists k$ such that $[a_k, b_k] \subset c$, which contradicts the assumption of finite covering.

6→6: Monotone bounded → Convergent

Proof: Let A be a nonempty real set, bounded above.

Suppose A has an element that is not its own upper bound. Let this element be a1.

A has an upper bound b1. Let c1 = (a1 + b1) / 2.

If c1 is an upper bound, let a2 = a1, b2 = c1; otherwise, let a2 = a1, b2 = b1.

Let c2 = (a2 + b2) / 2, ...

Thus, we get a sequence {bn} where each term is an upper bound of A. It is monotone decreasing, bounded below by a1. By the Monotone Convergence Theorem, {bn} converges. Let lim b_n = s.

∴ lim a_n = s.

∀x ∈ A ∀n x ≤ b_n. By monotonicity ∀x ∈ A x ≤ s.

∀ε > 0, ∴ lim a_n = s. ∃N ∈ N+, ∀n > N |a_n - s| < ε, a_n > s - ε.

Because a_{n+1} is not an upper bound, ∴ ∃x ∈ A x > a_{n+1} > s - ε.

∴ s is an upper bound.

4→1: Compactness (Bolzano-Weierstrass)

Proof: Suppose sequence {x_n} is bounded, a as the lower bound, b as the upper bound, but has no convergent subsequence.

∴ ∀A ∈ [a, b] ∃ε_A > 0, (A - ε_A, A + ε_A) contains only finite terms of {x_n}.

[a, b] ⊆ ∪ (A - ε_A, A + ε_A).

By the Compactness Theorem, there exist finite A1, ..., Am such that $[a, b] \subseteq \cup (A_i - \varepsilon_{A_i}, A_i + \varepsilon_{A_i})$.

∴ Each open interval $(A_i - \varepsilon_{A_i}, A_i + \varepsilon_{A_i})$ contains only finite terms of $\{x_n\}$.

∴ $\cup (A_i - \varepsilon_{A_i}, A_i + \varepsilon_{A_i})$ contains only finite terms of $\{x_n\}$.

∴ $[a, b]$ contains only finite terms of $\{x_n\}$.

1→4: Compactness (Bolzano-Weierstrass) → Finite Covering

Proof: With 3→4's proof method, by forming a sequence of intervals a_n, b_n, we can derive $\lim a_n = \lim b_n = c$.

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7→2: Completeness (Cauchy) → Monotonic and Bounded

Proof: Assume the sequence $\{a_n\}$ is monotonic increasing and bounded but divergent. By the Cauchy convergence criterion, $\exists\varepsilon>0, \forall N, \exists m,n>N, |a_m - a_n| > \varepsilon$. For $N=1$, $\exists m_1 > n_1$, $|a_{m_1} - a_{n_1}| > \varepsilon$. For $N=m_1$, $\exists m_2 > n_2 > m_1$, $|a_{m_2} - a_{n_2}| > \varepsilon$. For $N=m_2$, $\exists m_3 > n_3 > m_2$, $|a_{m_3} - a_{n_3}| > \varepsilon$. ... $\therefore$ $\{a_n\}$ is monotonic increasing. $\therefore$ $\forall k\in N^*$, $a_{m_k} > a_{n_k} + \varepsilon \geq a_{m_{k-1}} + \varepsilon > (a_{n_{k-1}} + \varepsilon) + \varepsilon = a_{n_{k-1}} + 2\varepsilon \geq a_{m_{k-2}} + 2\varepsilon > \dots > a_{m_1} + (k-1)\varepsilon$. $\because$ ε is given, k can be greater than ∞. $\therefore$ $\{a_{m_k}\}$ is unbounded, which contradicts the assumption.

1→5: Convergence (Bolzano) → Convergent

Proof: Let A be a bounded infinite set, $\forall x\in A$. $A\setminus\{x_1\}$ is a bounded infinite set, $\forall x_1\in A\setminus\{x_1\}$. $A\setminus\{x_1, x_2\}$ is a bounded infinite set, $\forall x_2\in A\setminus\{x_1, x_2\}$.

5→1: Convergent → Convergent (Bolzano)

Proof: Assume $\{x_n\}$ is a bounded sequence, thus has a convergent subsequence, let $\lim_{k\to\infty} x_{n_k} = a$. $\forall\varepsilon>0, \exists K, \forall k>K, |x_{n_k} - a| < \varepsilon$. Thus, a's ε neighborhood contains infinitely many points $x_{n_{k+1}}, x_{n_{k+2}}, \dots$

- A is a finite set, at this time $\{x_n\}$ has infinitely many equal terms, these terms form a convergent constant sequence.

- A is an infinite set, at this time A has a convergent point.

The text and formulas in the image are as follows:

Let a be a point in set A. Take any term of the sequence, denoted as x_n.

Let $\delta_2 = \min(\frac{1}{2}, |a - x_1|)$, in the δ_2-neighborhood of a, take the term of the sequence with index greater than n, denoted as x_{n_2}. Let $\delta_3 = \min(\frac{1}{3}, |a - x_{n_2}|)$, in the δ_3-neighborhood of a, take the term of the sequence with index greater than n_2, denoted as x_{n_3}...

This way, we obtain the subsequence $\{x_{n_k}\}$ converging to a. Because

$\forall \epsilon > 0, \exists N, \frac{1}{N} < \epsilon$

$\therefore \forall k > N, |x_{n_k} - a| < \frac{1}{k} < \epsilon$

4 → 5: Finite cover → point.

Proof: Suppose there is an unbounded infinite set $S \subset [-m, m]$. Since S has a limit point, it must be contained in $[-m, m]$.

Now, assume that $[-m, m]$ does not contain any points of S.

Then, $\forall x \in [-m, m]$, $\exists \delta_x > 0$, such that $U(x, \delta_x) \cap S$ is a finite set. Let

$H = \{U(x, \delta_x) \mid x \in [-m, m]\}$,

Then H is a cover of $[-m, m]$. By the finite cover theorem, $\exists n$

$S \subset [-m, m] \subset \cup_{i=1}^{n} U(x_i, \delta_i)$

$\therefore U(x_i, \delta_i) \cap S$ is a finite set $(i = 1, 2, \dots, n)$

$\therefore$ The above statement contradicts the assumption that S is an infinite set.

The formulas are:

$\delta_2 = \min\left(\frac{1}{2}, |a - x_1|\right)$

$\delta_3 = \min\left(\frac{1}{3}, |a - x_{n_2}|\right)$

$\forall k > N, |x_{n_k} - a| < \frac{1}{k} < \epsilon$

$\forall x \in [-m, m], \exists \delta_x > 0$, such that $U(x, \delta_x) \cap S$ is a finite set

$H = \{U(x, \delta_x) \mid x \in [-m, m]\}$

$S \subset [-m, m] \subset \cup_{i=1}^{n} U(x_i, \delta_i)$

$\therefore U(x_i, \delta_i) \cap S$ is a finite set $(i = 1, 2, \dots, n)$

$\therefore$ The above statement contradicts the assumption that S is an infinite set.

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Chapter 9: Properties of Continuous Functions on Closed Intervals

- **Boundedness - Extreme Value Theorem**

If a function $f(x)$ is continuous on the closed interval $[a, b]$, then $f(x)$ has a maximum and minimum value. That is, there exist points ξ and η in $[a, b]$ such that for all $x \in [a, b]$, $f(\xi) \leq f(x) \leq f(\eta)$.

Proof:

For any $\varepsilon > 0$, since $f(x) \in C[a, b]$, we have:

$$
\forall x_0 \in [a, b], \exists \delta_{x_0} > 0, \forall x \in (x_0 - \delta_{x_0}, x_0 + \delta_{x_0}) \cap [a, b].
$$

$$
|f(x) - f(x_0)| < 1, |f(x)| < |f(x_0)| + 1.
$$

$$
\bigcup_{x_0 \in [a, b]} (x_0 - \delta_{x_0}, x_0 + \delta_{x_0}) \text{ forms an open cover of } [a, b].
$$

$$
\text{There exist finitely many } x_0_1, x_0_2, \ldots, x_0_n \text{ such that }

\]

\[

[a, b] \subseteq \bigcup_{k=1}^n (x_0_k - \delta_{x_0_k}, x_0_k + \delta_{x_0_k}).

\]

\[

\therefore |f(x)| \leq \max_{1 \leq k \leq n} \{ |f(x_0_k)| + 1 \} \text{ is bounded.}

\]

Next, we prove that $f(x)$ has a minimum value.

By the completeness principle, $f(x)$ has a lower bound, let $m = \inf_{x \in [a,b]} f(x)$.

For any $n \in N^+$, there exists $x_n \in [a,b]$ such that $f(x_n) < m + \frac{1}{n}$.

Since $\{x_n\} \subseteq [a,b]$ is bounded, there exists a convergent subsequence $\{x_{n_k}\}$ with limit $\xi \in [a,b]$.

Since $f(x)$ is continuous on $[a,b]$,

\[

\lim_{k \to \infty} f(x_{n_k}) = f(\xi).

\]

\[

\therefore f(\xi) < m + \frac{1}{n}.

\]

\[

\lim_{k \to \infty} f(x_{n_k}) \leq \lim_{k \to \infty} (m + \frac{1}{n_k}) = m.

\]

\[

\therefore f(\xi) \leq m.

\]

Since m is the lower bound, $m \leq f(\xi)$.

$\therefore f(\xi) = m$. Thus, $f(x)$ has a minimum value.

- **Intermediate Value Theorem**

If $f(x)$ is continuous on the closed interval $[a, b]$ and $f(a) \neq f(b)$, then for any $\mu \in (f(a), f(b))$ or $(f(b), f(a))$, there exists $\xi \in (a, b)$ such that $f(\xi) = \mu$.

$\exists \xi \in (a, b), f(\xi) = \mu$.

Proof: Let $a_1 = a, b_1 = b$, then $c_1 = \frac{a_1+b_1}{2}$.

If $f(c_1) = 0$, take c_1, and we are done.

If $f(c_1) < 0$, then $a_2 = c_1, b_2 = b_1$.

If $f(c_1) > 0$, then $a_2 = a_1, b_2 = c_1$, and $c_2 = \frac{a_2+b_2}{2}$.

Or at some step, we find $f(c_2) = 0$, or we generate an interval $[a_n, b_n]$ satisfying $f(a_n) < 0, f(b_n) > 0$, by the interval套定理.

$\exists\xi = \lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n, f(\xi) = 0$, so $f(a_n) \le 0, f(b_n) \ge 0$.

$\therefore f(\xi) = 0$.

The proof of the zero point existence theorem, and the intermediate value theorem.

$\because f(x) \in C[a,b], \therefore g(x) = f(x) - \mu \in C[a,b]$.

$\therefore g(a)g(b) < 0$.

By the zero point existence theorem, there exists $\xi \in (a,b)$, $g(\xi) = 0$, so $f(\xi) = \mu$.

- Uniform Continuity Theorem (Heine-Borel)

If $f(x)$ is continuous on $[a,b]$, then $f(x)$ is uniformly continuous on $[a,b]$.

Proof: For $\epsilon > 0, \forall x_0 \in [a,b], \exists\delta_{x_0} > 0, x \in (x_0 - \delta_{x_0}, x_0 + \delta_{x_0}) \cap [a,b]$ such that $|f(x) - f(x_0)| < \frac{\epsilon}{3}$.

Then $[a,b] \subseteq \cup_{x_0\in[a,b]}(x_0 - \frac{\delta_{x_0}}{3}, x_0 + \frac{\delta_{x_0}}{3})$.

$\therefore \exists x_1, x_2, \dots, x_n, [a,b] \subseteq \cup_{k=1}^{n}(x_{0k} - \frac{\delta_{x_k}}{3}, x_{0k} + \frac{\delta_{x_k}}{3})$.

Let $\delta = \min_{1\le k\le n}(\frac{\delta_{x_k}}{3})$, when $|x - t| < \delta$, $\exists l, m \in \{1, 2, \dots, n\}$.

$x \in (x_{0l} - \frac{\delta_{x_l}}{3}, x_{0l} + \frac{\delta_{x_l}}{3})$, $t \in (x_{0m} - \frac{\delta_{x_m}}{3}, x_{0m} + \frac{\delta_{x_m}}{3})$.

Assume $\delta_{x_1} \leq \delta_{x_m}$, then $x_1 \in (x_{0m} - \delta_{x_m}, x_{0m} + \delta_{x_m})$.

Thus, $|f(x) - f(t)| \leq |f(x) - f(x_1)| + |f(t) - f(x_m)| + |f(x_1) - f(x_m)|$.

$\therefore$ $f(x)$ is uniformly continuous on $[a, b]$.

Therefore, $|f(x) - f(t)| \leq |f(x) - f(x_1)| + |f(t) - f(x_m)| + |f(x_1) - f(x_m)|$.

$\therefore$ $f(x)$ is uniformly continuous on $[a, b]$.

$\therefore$ $f(x)$ is uniformly continuous on $[a, b]$.

Riemann Integration

- Given a finite partition $T = \{x_0, x_1, \ldots, x_n\} \subset [a, b]$, satisfying $a = x_0 < x_1 < x_2 < \ldots < x_n = b$. Then partition T is a partition of the interval [a, b]. Each interval $[x_{k-1}, x_k]$ $(k = 1, \ldots, n)$ is a subinterval of [a, b]. $\Delta x_k = x_k - x_{k-1}$ is the length of the subinterval. $\lambda = \max\{\Delta x_k\}$ is the diameter of the partition T.

If f(x) is defined on [a, b], for any partition $T = \{x_0, x_1, \ldots, x_n\}$, and any point $\xi_k \in [x_{k-1}, x_k]$ $(k = 1, \ldots, n)$, if $\lim(\xi_k) \Delta x_k$ exists, then $f(x) \in R[a, b]$.

$\forall I \in R, \forall \varepsilon > 0, \exists \delta > 0, \forall T[a, b] = \{x_0, x_1, \ldots, x_n\}, \forall \xi_k \in [x_{k-1}, x_k]$, if $\lambda < \delta$, then $|\Sigma(\xi_k) \Delta x_k - I| < \varepsilon$. (Definition)

∴ Not integrable as:

$\forall I \in R, \exists \varepsilon > 0, \forall \delta > 0, \exists T[a, b] = \{x_0, x_1, \ldots, x_n\}, \exists \xi_k \in [x_{k-1}, x_k]$, if $\lambda < \delta$, then $|\Sigma(\xi_k) \Delta x_k - I| < \varepsilon$.

- Necessary condition for integrable function:

If $f(x) \in R[a, b]$, then f(x) is bounded on [a, b].

Proof: Since $f(x) \in R[a, b]$, for any partition T and any point $\xi_k \in [x_{k-1}, x_k]$, if the diameter $\lambda < \delta$, then

$|\Sigma(\xi_k) \Delta x_k - \int_a^b f(x)\, dx| < 1$.

∴ When $\lambda < \delta$, all integral sums are bounded.

Suppose f(x) is unbounded on [a, b]. Then for any partition T of [a, b] with any diameter less than δ, the partition f(x) must be unbounded in at least one subinterval. Suppose f(x) is unbounded in $[x_0, x_1]$. ∴ $\forall M > 0, \exists \xi_k \in [x_{k-1}, x_k]$ $(k = 2, 3, \ldots, n)$, such that $|f(\xi_k)| > |\Sigma(\xi_k) \Delta x_k| + M$.

$\therefore$ The integral sum and the integral in $\lambda < \delta$ are unbounded. $\therefore$ The integral sum and the integral in $\lambda < \delta$ are unbounded.

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- Let f(x) be bounded on [a, b]. T is a partition of [a, b]. Define:

$$M_k = \sup_{x\in[x_{k-1},x_k]}\{f(x)\},\quad m_k = \inf_{x\in[x_{k-1},x_k]}\{f(x)\},\quad k = 1,2,\dots,n$$

The upper sum of f(x) corresponding to partition T is:

$$S(f,T) = \sum_{k=1}^{n} M_k\Delta x_k$$

The lower sum of f(x) corresponding to partition T is:

$$s(f,T) = \sum_{k=1}^{n} m_k\Delta x_k$$

Since $m_k \le f(\xi_k) \le M_k$ (for partition T),

$$\sum_{k=1}^{n} m_k\Delta x_k \le \sum_{k=1}^{n} f(\xi_k)\Delta x_k \le \sum_{k=1}^{n} M_k\Delta x_k,\quad \xi_k \in [x_{k-1},x_k]$$

Thus,

$$s(f,T) \le \sum_{k=1}^{n} f(\xi_k)\Delta x_k \le S(f,T)$$

For the upper sum $S(f,T)$, we have:

$$\sum_{k=1}^{n} f(\xi_k)\Delta x_k \le S(f,T)$$

For any $\epsilon > 0$, since $M_k = \sup_{x\in[x_{k-1},x_k]}\{f(x)\}$, there exists $\xi_k \in [x_{k-1},x_k]$ such that $M_k - \frac{\epsilon}{2^n} < f(\xi_k)$. Thus,

$$\sum_{k=1}^{n} (M_k - \frac{\epsilon}{2^n})\Delta x_k < \sum_{k=1}^{n} f(\xi_k)\Delta x_k$$

This implies:

$$S(f, T) - \epsilon < \sum_{k=1}^{n} f(\xi_k)\Delta x_k$$

From (*) and (**), we have:

$$S(f, T) = \sup_{\xi_k \in [x_{k-1}, x_k]} \sum_{k=1}^{n} f(\xi_k)\Delta x_k$$

Similarly,

$$s(f, T) = \inf_{\xi_k \in [x_{k-1}, x_k]} \sum_{k=1}^{n} f(\xi_k)\Delta x_k$$

Let T_1 and T_2 be two partitions of [a, b] such that $T_1 \subseteq T_2$. Then $s(f, T_1) \leq s(f, T_2)$ and $S(f, T_2) \leq S(f, T_1)$.

Proof: Consider $T_1 = \{x_0, x_1, \ldots, x_n\}$ and let $t \in (x_0, x_1)$. Then T_2 is $a = x_0 < t < x_1 < x_2 < \ldots < x_n = b$.

Let $M_k = \sup_{x \in [x_{k-1}, x_k]}\{f(x)\}, k = 1, 2, \ldots, n, M' = \sup_{x \in [x_0, t]}\{f(x)\}$, $M'' = \sup_{x \in (t, x_1]}\{f(x)\}$.

Then $M_1 \geq M'$ and $M_1 \geq M''$.

Thus,

$$M'(t - x_0) + M''(x_1 - t) \leq M_1(t - x_0) + M_1(x_1 - t) = M_1(x_1 - x_0)$$

Therefore,

$$S(f, T_2) - S(f, T_1) = M'(t - x_0) + M''(x_1 - t) - M_1(x_1 - x_0) \leq 0$$

Hence,

$$S(f, T_2) \leq S(f, T_1)$$

Therefore,

$$s(f, T_1) \leq s(f, T_2)$$

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The text and formulas in the image are as follows:

If $T1$ and $T2$ are any two partitions of $[a, b]$, then $S(f, T1) \leq S(f, T2)$.

Proof: Let $T = T1 \cup T2$, then $T1 \subseteq T$, $T2 \subseteq T$. Therefore, $S(f, T1) \leq S(f, T)$, $S(f, T) \leq S(f, T2)$. Hence, $S(f, T1) \leq S(f, T2)$.

Combining the above two propositions, for any $T1$ and $T2$, $T = T1 + T2$, then $S(T1) \leq S(T) \leq S(T) \leq S(T2)$. Thus, the lower sum S has an upper bound, and the upper sum S has a lower bound. Therefore, there exists a definite integral $I = \int ab\ f(x)dx$, which is the definite integral of $f(x)$ over $[a, b]$.

Since the upper sum is the upper bound of the lower sum, for any $\varepsilon > 0$, there exists $T1$ such that $S(f, T1) < I + \varepsilon/2$. Let $\delta > 0$ be the smallest interval length in T, and T is a partition of $[a, b]$. Let $T2 = T \cup T1$. Then $S(f, T2) \leq S(f, T1)$. Therefore, $S(f, T2) \leq S(f, T)$.

Thus, $0 \leq S(f, T) - I = S(f, T) - S(f, T2) + S(f, T2) - I \leq S(f, T) - S(f, T2) + S(f, T2) - I < S(f, T) - S(f, T2) + \varepsilon/2$.

- When $T2 = T$, i.e., all the points of T are also points of T. $S(f, T) - S(f, T2) = 0$.
- When T2 has only one point t that is not a point of T. Suppose $t \in (x_{i-1}, x_i)$.

Therefore, $S(f, T) - S(f, T2) = M(x_i - x_{i-1}) - [M(t - x_{i-1}) + M(x_i - t)] \leq 2M(x_i - x_{i-1}) \leq 2M\delta$. Where $M = \sup_{x\in[a,b]} |f(x)|$.

Given $\delta = \min\{\varepsilon/4n, M, \delta\}$, then when T has a diameter $\lambda < \delta$, there is:

$0 \leq S(f, T) - S(f, T2) < \varepsilon/2$.

Therefore, $0 \leq S(f, T) - I < S(f, T) - S(f, T2) + \varepsilon/2 < \varepsilon/2 + \varepsilon/2 = \varepsilon$. Thus, $\lim S(f, T) = I$.

The formulas are:

- $S(f, T1) \leq S(f, T)$
- $S(f, T) \leq S(f, T2)$
- $S(f, T1) \leq S(f, T2)$
- $S(f, T1) \leq S(f, T2)$
- $S(f, T) - S(f, T2) = 0$
- $S(f, T) - S(f, T2) = M(x_i - x_{i-1}) - [M(t - x_{i-1}) + M(x_i - t)]$
- $S(f, T) - S(f, T2) \leq 2M(x_i - x_{i-1}) \leq 2M\delta$
- $S(f, T) - S(f, T2) \leq \varepsilon/2$
- $S(f, T) - S(f, T2) < \varepsilon/2$
- $S(f, T) - S(f, T2) + \varepsilon/2 < \varepsilon/2 + \varepsilon/2 = \varepsilon$
- $\lim S(f, T) = I$

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- Necessary and sufficient conditions for integrability: $I = \bar{I}$

Necessary condition: Suppose $f(x)$ is integrable on $[a, b]$. $\lim (\Sigma f(\xi_k)\Delta x_k) = I$

For any $\varepsilon > 0$, there exists $\delta > 0$. When the diameter of $T < \delta$, $|\Sigma f(\xi_k)\Delta x_k - I| < \varepsilon/2$.

For any partition T, there exists (ξ_k) $(k=1, 2, \ldots, n)$. Because the upper sum is the upper bound and the lower sum is the lower bound

$\Sigma f(\xi_k)\Delta x_k > S(f, T) - \varepsilon/2$

$|S(f, T) - I| \leq |S(f, T) - \Sigma f(\xi_k)\Delta x_k| + |\Sigma f(\xi_k)\Delta x_k - I| < \varepsilon$

$\lim S(f, T) = I$

Similarly, $\lim S(f, T) = I$. Therefore, $I = \bar{I}$

Sufficient condition: Suppose $\lim S(f, T) = \lim S(f, T)$, then $I = \bar{I}$

Since $S(f, T) \leq \Sigma f(\xi_k)\Delta x_k \leq S(f, T)$.

$I = \lim S(f, T) \leq \lim \Sigma f(\xi_k)\Delta x_k \leq \lim S(f, T) = \bar{I}$.

Since $I = \bar{I}$, $\lim \Sigma f(\xi_k)\Delta x_k = I = \bar{I}$. Thus, $f(x)$ is integrable on $[a, b]$.

Since $I = \bar{I}$, $\lim (S(f, T) - S(f, T)) = 0$. Therefore, $\lim \Sigma (M_k - m_k)\Delta x_k = 0$. Define $w_k = M_k - m_k$.

$\lim \Sigma w_k\Delta x_k = 0$.

- Commonly Integrable Functions

① $f(x)$ is continuous on $[a, b]$ implies $f(x)$ is integrable on $[a, b]$.

Proof: Since $f(x)$ is continuous on $[a, b]$, it is uniformly continuous on $[a, b]$.

For any $\varepsilon > 0$, there exists $\delta > 0$. When x' and $x'' \in [a, b]$ and $|x' - x''| < \delta$, $|f(x') - f(x'')| < \varepsilon/2$.

Therefore, for any partition T, the sum of $w_k \Delta x_k$ is zero.

$\omega_k = M_k - m_k = f(\xi_k) - f(\eta_k) < \varepsilon/2 - a$.

Thus, for any partition T, the sum of $w_k \Delta x_k$ is zero.

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- If $f(x)$ is bounded and has only a finite number of discontinuities in [a, b], then $f(x)$ is integrable in [a, b].

Proof: Assume there is only one discontinuity point c such that $a < c < b$.

Let $\varepsilon > 0$. Suppose $|f(c)| \leq M$, $\forall x \in [a, b]$.

Take $\varepsilon 1 = \varepsilon / (1 + 8M)$, $C1 = c - \varepsilon 1$, $C2 = c + \varepsilon 1$, $A = (C1, f(C1))$, $B = (C2, f(C2))$.

If $F(x) = \{f(x), x \in [a, c1] \cup [c2, b]; g(x), x \in (c1, c2)\}$, $g(x)$ is the line segment AB.

Then $F(x)$ is continuous in [a, b]. By the previous proposition, $F(x)$ is integrable in [a, b].

For any partition T, as long as $\lambda < \delta$, there is a sum of the form $\sum(w_k * \Delta x_k) < \varepsilon$.

- In $T = \{x_0, ..., x_n\}$ of the various intervals $[x_{k-1}, x_k]$, one class completely includes $[a, c1] \cup [c2, b]$.

Using the subscript 'k' to indicate those with intersections, and 'k" to indicate those without, the total length is less than 4ε.

$\therefore \sum(w_k * \Delta x_k) = \sum(w_k' * \Delta x_k') + \sum(w_k'' * \Delta x_k'') \leq \sum(w_k' * \Delta x_k') + \sum(w_k'' * \Delta x_k'') \leq \varepsilon + 2M * 4\varepsilon = \varepsilon + 8M * 4\varepsilon < 2\varepsilon$.

Thus, $\lim(\varepsilon \to 0) \sum(w_k * \Delta x_k) = 0$. The proof is similar for multiple discontinuities.

- If $f(x)$ is monotonic in [a, b], then $f(x)$ is integrable in [a, b].

Proof: Assume $f(x)$ is monotonic and non-constant in [a, b].

For any $\varepsilon > 0$, take $\delta = \varepsilon / (f(b) - f(a)) > 0$. For any partition T of [a, b], as long as $\lambda < \delta$, we have:

$0 < \sum(w_k * \Delta x_k) = \sum(f(x_k) - f(x_{k-1})) * \Delta x_k < \varepsilon$.

$= \varepsilon / (f(b) - f(a)) * \sum(f(x_k) - f(x_{k-1})) = \varepsilon.$

Thus, $\lim(\varepsilon \to 0) \sum(w_k * \Delta x_k) = 0.$

- Weighted Function Properties

- Linearity:

$$\int_a^b (\alpha f(x) + \beta g(x))dx = \alpha \int_a^b f(x)dx + \beta \int_a^b g(x)dx$$

Proof: For any $\alpha \in R$, any partition T of $[a, b]$, let ω_k^α, ω_k^β represent $\alpha f(x)$, $f(x)$ in the subinterval $[x_{k-1}, x_k]$. Then $\omega_k^\alpha = |\alpha|\omega_k^\beta$.

$\therefore \sum \ \omega_k^\alpha \Delta x_k = 0$

$\therefore \sum \ \omega_k^\alpha \Delta x_k = 0 \therefore \alpha f(x)$ is integrable.

Using ω_k, ω_k^β, ω_k^α to represent $f(x) + g(x)$, $f(x)$, $g(x)$ in the subinterval $[x_{k-1}, x_k]$. Then $\omega_k \le \omega_k^\beta + \omega_k^\alpha$

$\therefore \sum \ \omega_k^\alpha \Delta x_k = \sum \ \omega_k^\beta \Delta x_k = 0$

$\therefore \sum \ \omega_k^\alpha \Delta x_k = 0 \therefore f(x) + g(x)$ is integrable.

- Interval Additivity:

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$$

Proof: 1) $\Rightarrow$:

Let T_1 be any partition of $[a, c]$, T_2 be any partition of $[c, b]$. $T = T_1 \cup T_2$ is a partition of $[a, b]$.

$\sum_{k=1}^{n} \omega_k \Delta x_k + \sum_{k=n+1}^{m} \omega_k \Delta x_k = \sum_{k=1}^{m} \omega_k \Delta x_k$

$\sum_{k=1}^{n} f(x_k) \Delta x_k + \sum_{k=n+1}^{m} f(x_k) \Delta x_k = \sum_{k=1}^{m} f(x_k) \Delta x_k$

$\therefore \lambda(T) = \max\{\lambda(T_1), \lambda(T_2)\}$, and $f(x)$ is integrable on $[a, b]$.

$\therefore$ when $\lambda(T_1)$ and $\lambda(T_2)$ both tend to 0, $\lim_{\lambda(T)\to 0}\sum_{k=1}^{m}\omega_k\Delta x_k = 0$

$\therefore 0 \le \sum_{k=1}^{n}\omega_k\Delta x_k \le \sum_{k=1}^{m}\omega_k\Delta x_k, 0 \le \sum_{k=n+1}^{m}\omega_k\Delta x_k \le \sum_{k=1}^{m}\omega_k\Delta x_k$

$\therefore \lim_{\lambda(T)\to 0}\sum_{k=1}^{n}\omega_k\Delta x_k = \lim_{\lambda(T)\to 0}\sum_{k=1}^{m}\omega_k\Delta x_k = 0$

$\therefore \int_a^c f(x)dx + \int_c^b f(x)dx = \int_a^b f(x)dx$

- $\Leftarrow$:

Let $T = \{x_0, x_1, \dots, x_n\}$ be any partition of $[a, b]$. Suppose $c \in [x_{i-1}, x_i]$.

Then $x_0 < x_1 < \dots < x_{i-1} < c$ is a partition of $[a, c]$, and $c < x_i < \dots < x_n$ is a partition of $[c, b]$.

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$\exists M, \forall x \in [a,b], |f(x)| \leq M$, assume M is the diameter.

$\therefore 0 \leq \sum_{k=1}^{n} w_k \Delta x_k = \sum_{k=1}^{n} w_k \Delta x_k + w_i \Delta x_i + \sum_{k=i+1}^{n} w_k \Delta x_k$

$\leq [\sum_{k=1}^{i} (w_k \Delta x_k + w'(c - x_{i-1}))] + [\sum_{k=i+1}^{n} (w_k \Delta x_k + w''(x_i - c))] + 2M\Delta$

$\because f(x)$ is in $[a,c]$ and $[c,b]$ on the interval

$\therefore \int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$

(3) Absolute value function integrability: $|\int_a^b f(x)dx| \leq \int_a^b |f(x)|dx$

Proof: Let $T = \{x_0, x_1, \dots, x_n\}$ be any partition of $[a,b]$, then

$w_k = \sup_{x \in [x_{k-1}, x_k]} \{|f(x)| - |f(x)|\}$,

$\bar{w}_k = \sup_{x \in [x_{k-1}, x_k]} \{|f(x)| - |f(x)|\}$

Then $\bar{w}_k \leq w_k (k = 1, 2, \dots, n)$

$\therefore 0 \leq \sum_{k=1}^{n} \bar{w}_k \Delta x_k \leq \sum_{k=1}^{n} w_k \Delta x_k$

$\therefore \lim_{\Delta \to 0} \sum_{k=1}^{n} \bar{w}_k \Delta x_k = 0$

$\therefore |f(x)|$ is integrable on $[a,b]$.

According to $-|f(x)| \leq f(x) \leq |f(x)|$ and the integral comparison principle, we have

$-\int_a^b |f(x)|dx \leq \int_a^b f(x)dx \leq \int_a^b |f(x)|dx$

(4) Product function integrability: $f(x), g(x)$ are integrable on $[a,b]$, then $f(x)g(x)$ is also integrable on $[a,b]$.

Proof: Let T be any partition of $[a, b]$, then $w_k = \sup_{x \in [x_{k-1}, x_k]}\{|f(x)g(x) - f(x)g(x)|\}$,

M_k, m_k, w_k are the upper bound, lower bound, and range of $f(x)$ on $[x_{k-1}, x_k]$,

M_k, m_k, w_k are the upper bound, lower bound, and range of $g(x)$ on $[x_{k-1}, x_k]$.

$\therefore f(x), g(x) \geq 0$

$$\therefore w_k \le M_k\overline{M}_k - m_k\overline{m}_k$$

$$= M_k(\overline{M}_k - \overline{m}_k) + \overline{m}_k(M_k - m_k)$$

$$\le M_k w_k^f + \overline{M}_k w_k^f \quad \forall x \in [a,b],\ \text{if } |w_k| \le M, |g(x)| \le \overline{M}.$$

$$\therefore 0 \le \sum_{k=1}^{n} w_k \Delta x_k \le M \sum_{k=1}^{n} w_k^f \Delta x_k + \overline{M} \sum_{k=1}^{n} w_k^f \Delta x_k$$

$$\therefore \lim_{n\to\infty} \sum_{k=1}^{n} w_k^f \Delta x_k = \lim_{n\to\infty} \sum_{k=1}^{n} w_k^f \Delta x_k = 0$$

$$\therefore \lim_{n\to\infty} \sum_{k=1}^{n} w_k \Delta x_k = 0.$$

When $f(x)$ and $g(x)$ are of opposite signs, let $f^+(x) = \frac{1+f(x)}{2}$, $f^-(x) = \frac{1-f(x)}{2}$, $g^+(x) = \frac{1+g(x)}{2}$, $g^-(x) = \frac{1-g(x)}{2}$. Then $f^+(x), f^-(x), g^+(x), g^-(x) \ge 0$.

$\because$ $f(x), g(x)$ are integrable, $\therefore$ $f^+(x), f^-(x), g^+(x), g^-(x)$ are all integrable.

$\therefore f(x)g(x) = [f^+(x) - f^-(x)][g^+(x) - g^-(x)]$

$= f^+(x)g^+(x) - f^+(x)g^-(x) + f^-(x)g^+(x) + f^-(x)g^-(x)$

$\therefore$ each term is integrable, $\therefore$ $f(x)g(x)$ is integrable.

(5) First Mean Value Theorem: If $f(x)$ is continuous on $[a,b]$, $g(x)$ is integrable and non-zero on $[a,b]$, then there exists $\xi \in (a,b)$ such that $\int_a^b f(x)g(x)dx = f(\xi)\int_a^b g(x)dx$.

Proof: Since $f(x)$ is continuous on $[a,b]$, there exists $x_1, x_2 \in [a,b]$, $\forall x \in [a,b]$ such that $m = f(x_1) \le f(x) \le f(x_2) = M$.

$\because$ $g(x)$ is non-zero, $\therefore$ it is not zero. Then $mg(x) \le f(x)g(x) \le Mg(x)$.

$\therefore m\int_a^b g(x)dx \le \int_a^b f(x)g(x)dx \le M\int_a^b g(x)dx$.

When $\int_a^b g(x)dx = 0$, by the above formula, $\int_a^b f(x)g(x)dx = f(\xi)\int_a^b g(x)dx = 0$.

$\therefore$ when $\int_a^b g(x)dx > 0$, let $\mu = \dfrac{\int_a^b f(x)g(x)dx}{\int_a^b g(x)dx}$, then $m \le \mu \le M$.

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The text and formulas in the image are as follows:

If $m < \mu < M$, according to the Intermediate Value Theorem, there exists $\xi \in (x_1, x_2) \subset (a, b)$ such that $f(\xi) = \mu$. That is, $\int_a^b f(x)g(x)dx = f(\xi)\int_a^b g(x)dx$.

If $\mu = m$ or $\mu = M$, it contradicts $\mu = m$. Since $\int_a^b g(x)dx > 0$, there exists $h > 0$ such that $\int_{a+h}^{b-h} g(x)dx > 0$. If $f(x) - \mu = f(x) - m$ in (a, b) is always greater than 0, then there exists $m_0 > 0$. For $x \in [a + h, b - h]$, we have $f(x) - \mu \geq m_0 > 0$.

$$
\therefore \int_a^b f(x) g(x) dx - \mu \int_a^b g(x) dx = \int_a^b (f(x) - \mu) g(x) dx = \int_a^b (f(x) - m) g(x) dx
$$

$$
\geq \int_{a+h}^{b-h} (f(x) - \mu) g(x) dx \geq m_0 \int_{a+h}^{b-h} g(x) dx > 0
$$

This contradicts $\int_a^b f(x)g(x)dx = \mu\int_a^b g(x)dx$. Therefore, $\exists\xi \in (a, b), f(\xi) = \mu$.

The Second Integral Mean Value Theorem: If $f(x)$ is monotonic in $[a, b]$, then there exists $\xi \in [a, b]$ such that

$$
\int_a^b f(x) g(x) dx = f(a) \int_a^b g(x) dx + f(b) \int_a^b g(x) dx.
$$

The text discusses the Intermediate Value Theorem and the Second Integral Mean Value Theorem in the context of definite integrals. It proves the existence of a point ξ within the interval (a, b) where the function $f(x)$ equals a given value μ, and it also discusses the conditions under which the integral of a product of two functions can be expressed in terms of the function values at the endpoints of the interval.

Chapter 11. Uniform Convergence of Function Series

1. Definition of Uniform Convergence:

A sequence of functions $\{f_n(x)\}$ defined on I converges uniformly to $f(x)$ if for all $\varepsilon > 0$, there exists an N such that for all $n > N$ and all x in I, $|f_n(x) - f(x)| < \varepsilon$.

2. Necessary and Sufficient Conditions:

Necessary condition: For all $\varepsilon > 0$, there exists an N such that for all $n > N$ and all x in I, $|f_n(x) - f(x)| < \varepsilon$. Cauchy condition.

Sufficient condition: $\lim \sup |S(x) - S_n(x)| = 0$. Proof:

Necessary condition: If $\{f_n(x)\}$ converges uniformly to $f(x)$, then for all $\varepsilon > 0$, there exists an N such that for all $n > N$ and all x in I, $|f_n(x) - f(x)| < \varepsilon$. Let $p > 0$, then $n + p > N$. Thus, $|S_{n+p}(x) - S_n(x)| < \varepsilon/2$. Since $|S_{n+p}(x) - S_n(x)| \leq |S_{n+p}(x) - S(x)| + |S(x) - S_n(x)| < \varepsilon/2 + \varepsilon/2 = \varepsilon$. Sufficient condition: If $\{S_n(x)\}$ converges to $S(x)$, then for all $\varepsilon > 0$, there exists an N such that for all $n > N$ and all x in I, $|S(x) - S_n(x)| < \varepsilon$. Let $p \to \infty$, then $|S(x) - S_n(x)| < \varepsilon$. Thus, $\{f_n(x)\}$ converges uniformly to $f(x)$. Proof by Cauchy convergence criterion.

3. Weierstrass Test:

The Weierstrass test states that if $\{f_n(x)\}$ is a sequence of functions defined on I such that for all x in I, there exists a constant M_n such that $|f_n(x)| \leq M_n$ and the series $\sum M_n$ converges, then the series $\sum f_n(x)$ converges uniformly on I.

The note discusses the convergence of series and functions. It includes theorems and proofs related to uniform convergence, monotonicity, and the integral test. The text also covers the properties of uniformly convergent series and functions, including continuity. The proofs involve epsilon-delta arguments and inequalities. The mathematical content is dense and technical, focusing on advanced calculus concepts.

The note discusses the convergence of series and the properties of functions. It states that for any x in the interval I, if the absolute value of the difference between f(x) and f_n(x) is less than delta, then the series converges. It also mentions that the series of functions is uniformly convergent on I. The note further explains that the partial sums of the series are continuous on I. It concludes by stating that the series is uniformly convergent on I. The note also discusses the concept of improper integrals and the conditions under which they can be evaluated. It states that if a function is continuous on the interval [a, b], then the improper integral of the function can be evaluated. The note also discusses the concept of the integral of a series and the conditions under which it can be evaluated. It states that if the series of functions is uniformly convergent on the interval [a, b], then the integral of the series can be evaluated. 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It states that if the series of functions is uniformly convergent on the interval [a, b], then the integral of the series can be evaluated. The note also discusses the

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The function f(x) is differentiable on [a, b]. And

$$\frac{d}{dx}[\lim_{n\to\infty} f_n(x)] = \lim_{n\to\infty} \frac{d}{dx} f_n(x)$$

.

Let u_n(x) ∈ C^1[a, b]. If u_n(x) converges uniformly on [a, b], and if the sequence u_n(x) converges uniformly on [a, b], then u_n(x) is differentiable on [a, b]. And

$$\frac{d}{dx}[\lim_{n\to\infty} u_n(x)] = \lim_{n\to\infty} \frac{d}{dx} u_n(x)$$

.

Proof: Suppose {f_n(x)} converges uniformly to g(x), and g(x) is continuous. Then g(x) is continuous on [a, x] for x ∈ (a, b). Therefore, f_n(x) = f_n(a) + ∫_a^x f_n'(t) dt, x ∈ [a, b].

Both sides of the equation are continuous. By the theorem of continuous functions, f(x) = f(a) + ∫_a^x g(t) dt. Since g(x) is continuous, f'(x) = g(x).

Thus,

$$\frac{d}{dx}[\lim_{n\to\infty} f_n(x)] = \lim_{n\to\infty} \frac{d}{dx} f_n(x)$$

.

Since the partial sum of the series {S_n(x)} converges uniformly on [a, b], and {S_n(x)} converges uniformly on [a, b], then S(x) is differentiable. Therefore, S'(x) = ∑_{n=1}^∞ S_n'(x).

Thus,

$$\frac{d}{dx}[\lim_{n\to\infty} S_n(x)] = \lim_{n\to\infty} [\frac{d}{dx} S_n(x)] = \lim_{n\to\infty} [\sum_{k=1}^{n} u_k(x)] = \sum_{k=1}^{\infty} \frac{d}{dx} u_k(x)$$

•

(To be continued)